Lecture 1: Introduction to Entity Relationship (ER) model

Dr Tran Duc Minh

Faculty of Data Science and Artificial Intelligence
College of Technology, National Economics University
Email: minhdt@neu.edu.vn

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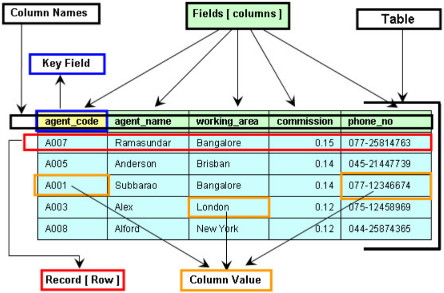
Outline

Overview

- Conceptual models are used in the conceptual design of databases.
- The Entity Relationship Model (ERM) is a conceptual model used to understand and design the data requirements of an organization.
- The Entity Relationship Diagram (ERD) is a graphical representation of the ERM. It helps to visualize the database structure, making it easier to understand how data is organized and how different pieces of data are related.
- ERDs depict the database's main components:
  - Entities
  - Attributes
  - Relationships

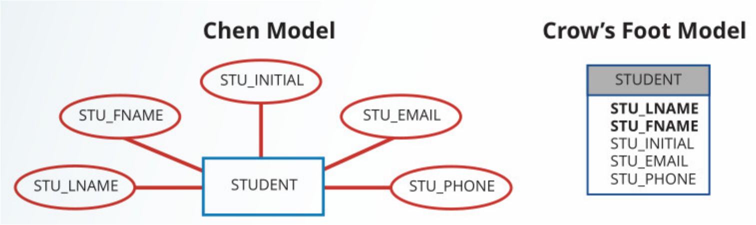
Entities

- An **entity** is an object of interest to the end user.
- An entity actually refers to the **entity set** and not to a single entity occurrence.
- An entity in the ERM corresponds to a **table** in the relational environment.
- The ERM refers to a **table row** as an **entity instance** or **entity occurrence**.



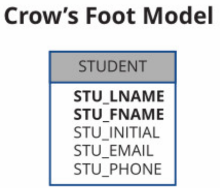
Attributes

- **Attributes** are characteristics of entities.
- For example, the STUDENT entity includes the attributes STU_LNAME, STU_FNAME, STU_INITIAL, STU_EMAIL and STU_PHONE



Required and Optional Attributes

- A **required attribute** is an attribute that must have a value.
- An **optional attribute** is an attribute that does not require a value.



- For example: The two boldfaced attributes in the Crow's Foot notation indicate that data entry will be required. The remaining attributes are not presented in boldface in the entity box, these are optional attributes.

Domains

- Attributes have a domain. A **domain** is the set of possible values for a given attribute.
- For example:
  - The domain for a grade point average (GPA) attribute is written (0, 10) because the lowest possible GPA value is 0 and the highest possible value is 10.
  - The domain for a gender attribute consists of only two possibilities: Male or Female.
  - The domain for a person's date of birth attribute consists of all dates that fit in a range.

Identifiers (Primary Keys)

- The ERM uses **identifiers** which are one or more attributes that uniquely identify each *entity instance*.
- In the relational model, entities are mapped to tables, and the entity identifier is mapped as the table's **primary key (PK)**.
- Identifiers are underlined in the ERD
  - IMAGE OF the Crow's Foot notation WITH UNDERLINE OF KEY (if possible)
  - Key attributes are also underlined in a frequently used shorthand notation for the table structure, called a **relational schema**, that uses the following format:

    CAR (**CAR_VIN**, MOD_CODE, CAR_YEAR, CAR_COLOR)

    Each car is identified by a unique Vehicle Identification Number, or CAR_VIN.

Composite Identifiers

- A **composite identifier** is a primary key composed of more than one attribute.
- For example: Each CLASS entity instance (occurrence) is identified using the composite primary key of CRS_CODE, CLASS_SECTION, and CLASS_SECTION.
  CLASS (**CLASS_CODE**, **CRS_CODE**, **CLASS_SECTION**, CLASS_TIME, ROOM_CODE, PROF_NUM)

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

Composite and Simple Attributes

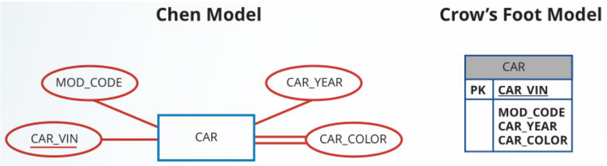
- A **composite attribute**, *not to be confused with a composite key*, is an attribute that can be further subdivided to yield additional attributes.
  - For example: A phone number such as 615-898-2368 may be divided into an area code (615), an exchange number (898), and a four-digit code (2368).
- A **simple attribute** is an attribute that cannot be subdivided.
  - For example: Age, sex, and marital status would be classified as simple attributes.

Single-Valued Attributes

- A **single-valued attribute** is an attribute that can have only a single value.
  - *A single-valued attribute is not necessarily a simple attribute.*
- For example
  - A part's serial number (such as SE-08-02-189935) is single-valued, but it is a composite attribute because it can be subdivided into the region in which the part was produced (SE), the plant within that region (08), the shift within the plant (02), and the part number (189935).

Multivalued Attributes

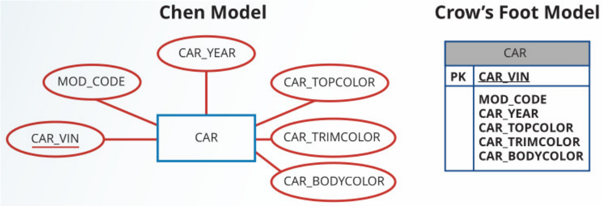
- **Multivalued attributes** are attributes that can have many values for a single entity instance.
- For example
  - An EMP_DEGREE attribute might store the string "BBA, MBA, PHD" to indicate three different degrees held.
- In the Chen ERM, multivalued attributes are shown by a double line connecting the attribute to the entity. The Crow's Foot notation does not identify multivalued attributes.



Implementing Multivalued Attributes

If multivalued attributes exist, the designer must decide on one of two possible courses of action:

1. Within the original entity, create several new attributes, one for each component of the original multivalued attribute.
    - For example: the CAR entity's attribute CAR_COLOR can be split to create the new attributes CAR_TOPCOLOR, CAR_BODYCOLOR, and CAR_TRIMCOLOR, which are then assigned to the CAR entity.



    - The downside to this approach is that new changes in the environment will never create a situation where an instance would have more values than before.

Implementing Multivalued Attributes

If multivalued attributes exist, the designer must decide on one of two possible courses of action:

2. Create a new entity composed of the original multivalued attribute's components. Then, this new entity is related to the original entity in a 1:M relationship.

   For example:

   - Creating an entity named CAR_COLOR allows the designer to define color for different sections of the car.

Section	Color
Top	White
Body	Blue
Trim	Gold
Interior	Blue

Implementing Multivalued Attributes

If multivalued attributes exist, the designer must decide on one of two possible courses of action:

2. Create a new entity composed of the original multivalued attribute's components. Then, this new entity is related to the original entity in a 1:M relationship.

   For example:
   - Then, this new CAR_COLOR entity is related to the original CAR entity in a 1:M relationship.

Implementing Multivalued Attributes

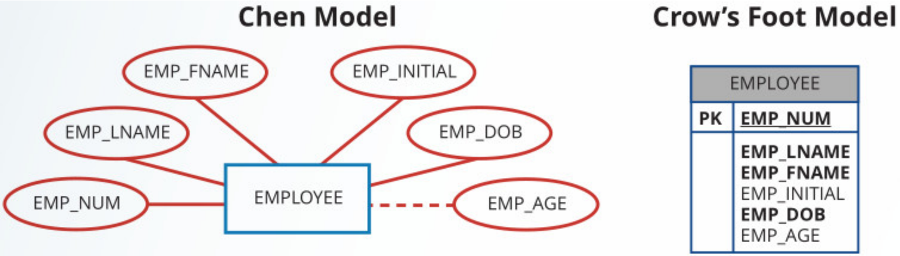
If multivalued attributes exist, the designer must decide on one of two possible courses of action:

2. Create a new entity composed of the original multivalued attribute's components. Then, this new entity is related to the original entity in a 1:M relationship.
    - Creating a new entity in a 1:M relationship with the original entity yields several benefits: it is a more flexible, expandable solution, and it is compatible with the relational model.

Derived Attributes

- A **derived attribute** is an attribute whose value is calculated (derived) from other attributes.
- The derived attribute need not be physically stored within the database; instead, it can be derived by using an algorithm.
- For example
  - An employee's age, EMP_AGE, may be found by computing the integer value of the difference between the current date and the EMP_DOB.
- A derived attribute is indicated in the Chen notation by a dashed line that connects the attribute and the entity. The Crow's Foot notation does not have a method for distinguishing the derived attribute from other attributes.

Derived Attributes



Derived Attributes

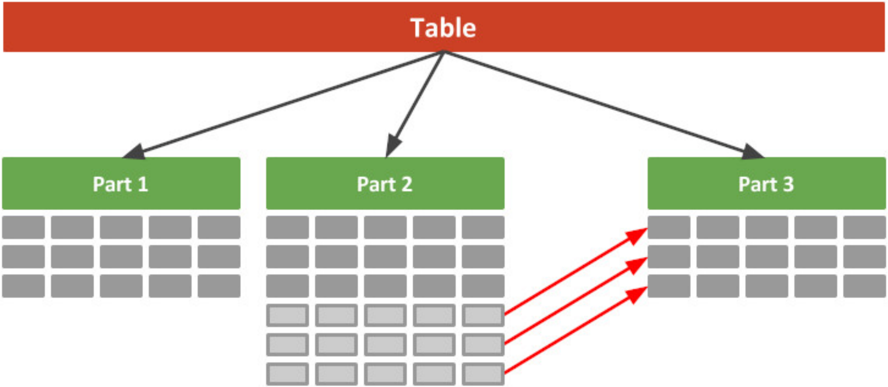
- Derived attributes are sometimes referred to as **computed attributes**.
- The decision to store derived attributes in database tables depends on the processing requirements and the constraints placed on a particular application. The designer should be able to balance the design in accordance with such constraints.
- Advantages and disadvantages of storing derived attributes

	Derived Attribute	
	Stored	**Not Stored**
Advantage	Saves CPU processing cycles Saves data access time Data value is readily available Can be used to keep track of historical data	Saves storage space Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	Uses CPU processing cycles Increases data access time Adds coding complexity to queries

Why should a larger table within a database be divided?

employer_id	skill	name	telephone_numbers	postal_code
105	typing	John Doe	028 90 23 10 23 0494 54 60 23 12	BT5 6AZ
105	cleaning	John Doe	028 90 23 10 23 0494 54 60 23 12	BT5 6AZ
213	cleaning	Jane Doe	028 12 34 12 34	BT4 2RE
213	catering	Jane Doe	028 12 34 12 34	BT4 2RE
213	carving	Jane Doe	028 12 34 12 34	BT4 2RE
176	carving	Thom Smith	0495 23 43 88 90	BT3 7DD

Why should a larger table within a database be divided?



Relationships

- A **relationship** is an association between entities.
- The entities that participate in a relationship are also known as **participants**, and each relationship is identified by a name that describes the relationship.
- The **relationship name** is an active or passive verb.
  - For example: a STUDENT takes a CLASS, a PROFESSOR teaches a CLASS, a DIVISION is managed by an EMPLOYEE, and an AIRCRAFT is flown by a CREW.

Relationships

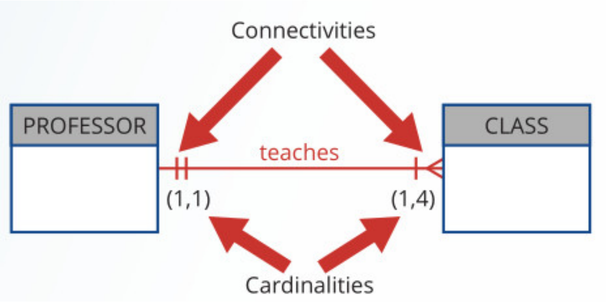
- Relationships between entities always operate in both directions.
- For example 1, to define the relations hip between the entities named CUSTOMER and INVOICE, we would specify that:
  - A CUSTOMER may generate many INVOICEs.
  - Each INVOICE is generated by one CUSTOMER.
    Because we know both directions of the relationship between CUSTOMER and INVOICE, it is easy to see that this relationship can be classified as 1:M.

Relationships

- The relationship classification is difficult to establish if we know only one side of the relationship.
- For example 2, if we specify that:
  - A DIVISION is managed by one EMPLOYEE.
    We don't know if the relationship is 1:1 or 1:M. Therefore, we should ask the question "Can an employee manage more than one division?"
  - If the answer is yes, the relationship is 1:M
  - If the answer is no, the relationship is 1:1

Connectivity and Cardinality

- The term **connectivity** is used to describe the classification of the relationship between entities. Classifications include 1:1, 1:M, and M:N.
- **Cardinality** expresses the minimum and maximum number of entity occurrences associated with one occurrence of the related entity.
    - In the ERD, cardinality is indicated by placing the appropriate numbers beside the entities using the format (x,y).



Existence Dependence

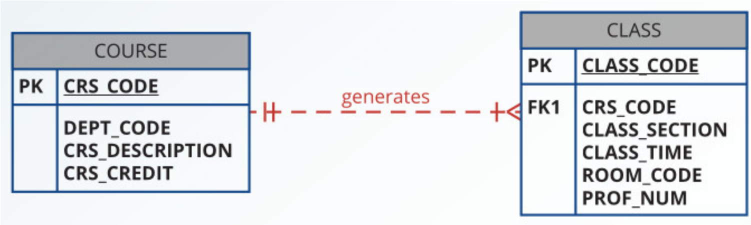
- An entity is said to be **existence-dependent** if its existence depends on one or more other entities.
  - An entity is existence-dependent if it has a mandatory foreign key (a foreign key attribute that cannot be null).
  - For example: If an employee wants to claim one or more dependents for tax-withholding purposes. The DEPENDENT entity is clearly existence-dependent on the EMPLOYEE entity because it is impossible for the dependent to exist apart from the EMPLOYEE in the database.

Existence Independence

- If an entity can exist apart from all of its related entities, then it is **existence-independent**, and it is referred to as a **strong entity** or **regular entity**.
- For example: In a university database, we have two entities, STUDENT and COURSE.
  - A STUDENT entity can exist independently of the COURSE entity because a student can be admitted to the university and exist in the database even if they haven't enrolled in any courses yet.
  - A COURSE entity can also exist independently of the STUDENT entity. The university can offer a course regardless of whether any students have enrolled in it.

Weak (Non-Identifying) Relationships

- The concept of **relationship strength** is based on how the primary key of a related entity is defined.
- A **weak relationship**, also known as a **non-identifying relationship**, exists if the primary key of the related entity does not contain a primary key component of the parent entity.
- The Crow's Foot notation depicts a weak relationship by placing a dashed relationship line between the entities.
    - For example:



Weak (Non-Identifying) Relationships

Table name: COURSE

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
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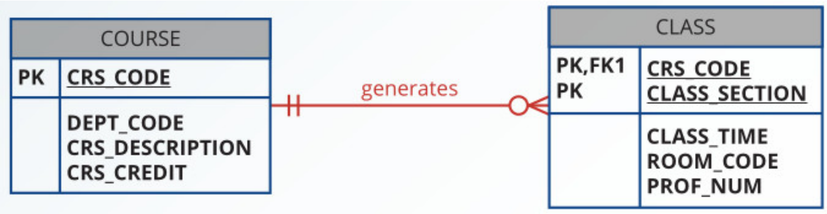
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

Strong (Identifying) Relationships

- A **strong (identifying) relationship** exists when the primary key of the related entity contains a primary key component of the parent entity.
- The Crow's Foot notation depicts the strong (identifying) relationship with a solid line between the entities
  - For example:



Strong (Identifying) Relationships

Table name: COURSE

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

Table name: CLASS

CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
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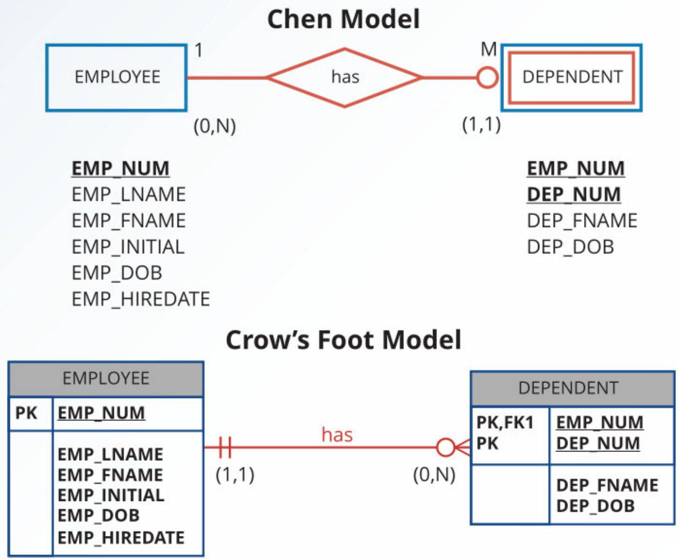
Weak Entities

- A **weak entity** is one that meets two conditions:
  1. The entity is existence-dependent; it cannot exist without the entity with which it has a relationship.
  2. The entity has a primary key that is partially or totally derived from the parent entity in the relationship.
- For example, a company insurance policy insures an employee and any dependents. An EMPLOYEE might or might not have a DEPENDENT, but the DEPENDENT must be associated with an EMPLOYEE. Moreover, the DEPENDENT cannot exist without the EMPLOYEE. DEPENDENT is the weak entity in the relationship "EMPLOYEE has DEPENDENT."

Weak Entities

- A strong (identifying) relationship indicates that the related entity is weak.
- The Chen notation identifies the weak entity by using a double-walled entity rectangle. The Crow's Foot notation uses the relationship line and the PK/FK designation to indicate whether the related entity is weak.

Weak Entities



Weak Entities

A Weak Entity in a Strong Relationship

Table name: EMPLOYEE **Database name: Ch04_ShortCo**

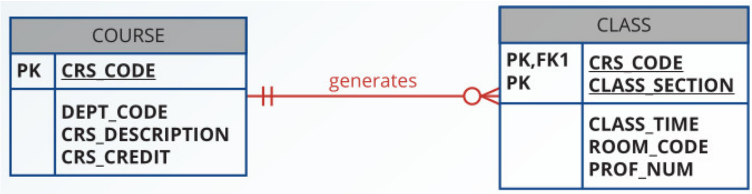
EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
1001	Callifante	Jeanine	J	12-Mar-64	25-May-97
1002	Smithson	William	K	23-Nov-70	28-May-97
1003	Washington	Herman	H	15-Aug-68	28-May-97
1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
1005	Johnson	Melanie		28-Sep-66	20-Dec-98
1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

Table name: DEPENDENT

EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
1001	1	Annelise	05-Dec-97
1001	2	Jorge	30-Sep-02
1003	1	Suzanne	25-Jan-04
1006	1	Carlos	25-May-01
1008	1	Michael	19-Feb-95
1008	2	George	27-Jun-98
1008	3	Katherine	18-Aug-03

Relationship Participation

- **Optional participation** means that one entity occurrence does not require a corresponding entity occurrence in a particular relationship.
- For example: In the "COURSE generates CLASS" relationship, an entity occurrence (row) in the COURSE table does not necessarily require the existence of a corresponding entity occurrence in the CLASS table. Therefore, the CLASS entity is considered to be optional to the COURSE entity.
- In Crow's Foot notation, an optional relationship between entities is shown by drawing a small circle (O) on the side of the optional entity.



Relationship Participation

- **Mandatory participation** means that one entity occurrence requires a corresponding entity occurrence in a particular relationship.
  - For example: An EMPLOYEE works in a DIVISION (A person cannot be an employee without being assigned to a company's division.)
- If the mandatory participation is depicted graphically, it is typically shown as a small hash mark across the relationship line, similar to the Crow's Foot depiction of a connectivity of 1.

Crow's Foot Symbols

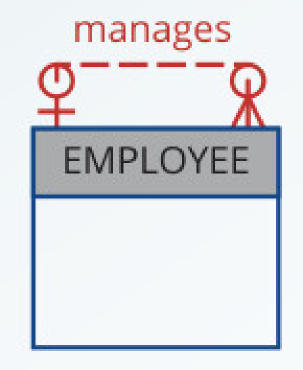
Symbol	Cardinality	Comment
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

Relationship Degree

- A **relationship degree** indicates the number of entities or participants associated with a relationship.
- A **unary relationship** exists when an association is maintained within a single entity.
  - For example: An EMPLOYEE might manage another EMPLOYEE.
- A **binary relationship** exists when two entities are associated.
  - For example: PROFESSOR teaches CLASS.
- A **ternary** exists when three entities are associated.
  - For example: A DOCTOR prescribes a DRUG for a PATIENT.

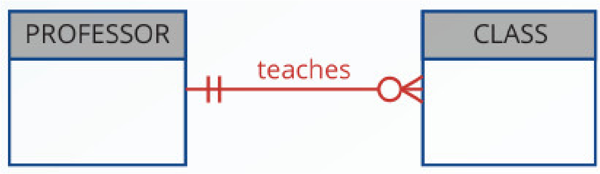
Unary Relationships

- For example: An employee within the EMPLOYEE entity is the manager for one or more employees within that entity.



Binary Relationships

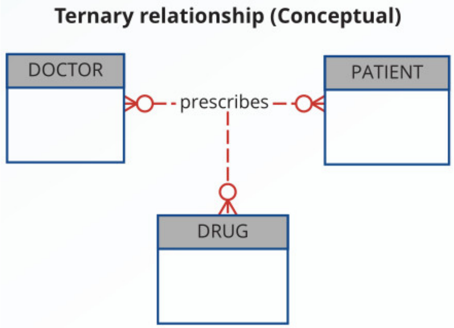
- A binary relationship exists when two entities are associated in a relationship.
- In fact, to simplify the conceptual design, most higher-order (ternary and higher) relationships are decomposed into appropriate equivalent binary relationships whenever possible.
- For example: "A PROFESSOR teaches one or more CLASSes" represents a binary relationship.



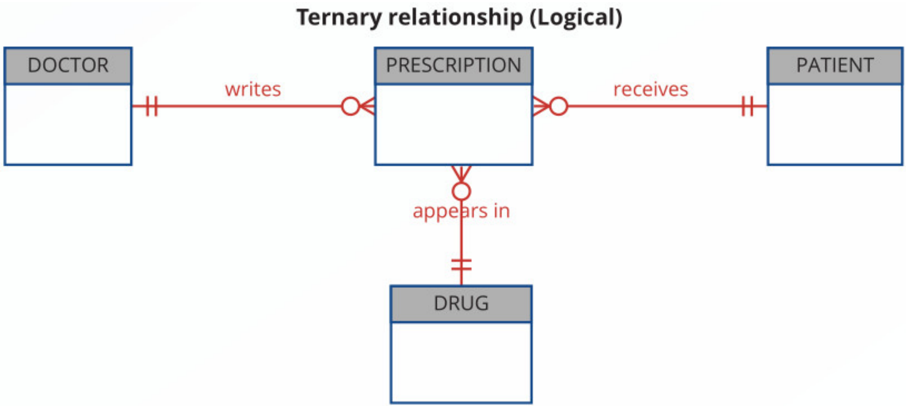
Ternary and Higher-Order Relationships

- A ternary relationship implies an association among three different entities.

  For example: A DOCTOR prescribes a DRUG for a PATIENT.
  - A DOCTOR writes one or more PRESCRIPTIONs.
  - A PATIENT may receive one or more PRESCRIPTIONs.
  - A DRUG may appear in one or more PRESCRIPTIONs.



Ternary and Higher-Order Relationships



Ternary and Higher-Order Relationships

The Implementation of a Ternary Relationship

Database name: Ch04_Clinic

Table name: DRUG

DRUG_CODE	DRUG_NAME	DRUG_PRICE
AF15	Afgapan-15	25.00
AF25	Afgapan-25	35.00
DRO	Droalene Chloride	111.89
DRZ	Druzocholar Cryptolene	18.99
KO15	Koliabar Oxyhexalene	65.75
OLE	Oleander-Drizapan	123.95
TRYP	Tryptolac Heptadimetric	79.45

Table name: PATIENT

PAT_NUM	PAT_TITLE	PAT_LNAME	PAT_FNAME	PAT_INITIAL	PAT_DOB	PAT_AREACODE	PAT_PHONE
100	Mr.	Kolmycz	George	D	15-Jun-1942	615	324-5456
101	Ms.	Lewis	Rhonda	G	19-Mar-2005	615	324-4472
102	Mr.	Vandam	Rhett		14-Nov-1958	901	675-8993
103	Ms.	Jones	Anne	M	16-Oct-1974	615	898-3456
104	Mr.	Lange	John	P	08-Nov-1971	901	504-4430
105	Mr.	Williams	Robert	D	14-Mar-1975	615	890-3220
106	Mrs.	Smith	Jeanine	K	12-Feb-2003	615	324-7883
107	Mr.	Diante	Jorge	D	21-Aug-1974	615	890-4567
108	Mr.	Wiesenbach	Paul	R	14-Feb-1966	615	897-4358
109	Mr.	Smith	George	K	18-Jun-1961	901	504-3339
110	Mrs.	Genkazi	Leighla	W	19-May-1970	901	569-0093
111	Mr.	Washington	Rupert	E	03-Jan-1966	615	890-4925
112	Mr.	Johnson	Edward	E	14-May-1961	615	898-4387
113	Ms.	Smythe	Melanie	P	15-Sep-1970	615	324-9006
114	Ms.	Brandon	Marie	G	02-Nov-1932	901	882-0845
115	Mrs.	Saranda	Hermine	R	25-Jul-1972	615	324-5505
116	Mr.	Smith	George	A	08-Nov-1965	615	890-2984

Table name: DOCTOR

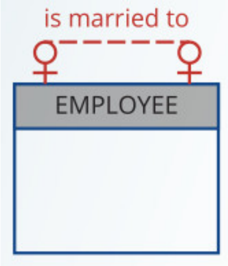
DOC_ID	DOC_LNAME	DOC_FNAME	DOC_INITIAL	DOC_SPECIALTY
29827	Sanchez	Julio	J	Dermatology
32445	Jorgensen	Annelise	G	Neurology
33456	Korenski	Anatoly	A	Urology
33989	LeGrande	George		Pediatrics
34409	Washington	Dennis	F	Orthopaedics
36221	McPherson	Katye	H	Dermatology
36712	Dreifag	Herman	G	Psychiatry
38995	Minh	Tran		Neurology
40004	Chin	Ming	D	Orthopaedics
40028	Feinstein	Denise	L	Gynecology

Table name: PRESCRIPTION

DOC_ID	PAT_NUM	DRUG_CODE	PRES_DOSAGE	PRES_DATE
32445	102	DRZ	2 tablets every four hours -- 50 tablets total	12-Nov-22
32445	113	OLE	1 teaspoon with each meal -- 250 ml total	14-Nov-22
34409	101	KO15	1 tablet every six hours -- 30 tablets total	14-Nov-22
36221	109	DRO	2 tablets with every meal -- 60 tablets total	14-Nov-22
38995	107	KO15	1 tablet every six hours -- 30 tablets total	14-Nov-22

Recursive Relationships

- A **recursive relationship** is one in which a relationship can exist between occurrences of the same entity set. Such a condition is found within a unary relationship.
- For example:
  - A 1:1 unary relationship may be expressed by "an EMPLOYEE may be married to one and only one other EMPLOYEE."



Recursive Relationships

- For example: The 1:1 Recursive Relationship "EMPLOYEE is married to EMPLOYEE"

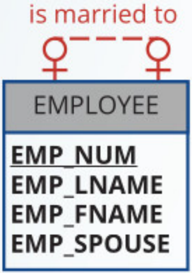
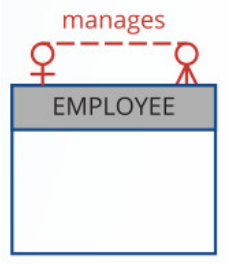


Table name: EMPLOYEE_V1

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_SPOUSE
345	Ramirez	James	347
346	Jones	Anne	349
347	Ramirez	Louise	345
348	Delaney	Robert	
349	Shapiro	Anton	346

Recursive Relationships

- A **recursive relationship** is one in which a relationship can exist between occurrences of the same entity set. Such a condition is found within a unary relationship.
- For example:
  - A 1:M unary relationship can be expressed by "an EMPLOYEE may manage many EMPLOYEEs, and each EMPLOYEE is managed by one EMPLOYEE."



Recursive Relationships

- For example: Implementation of the 1:M Recursive Relationship "EMPLOYEE manages EMPLOYEE"

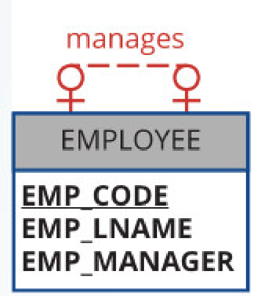
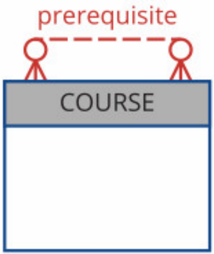


Table name: EMPLOYEE_V2

EMP_CODE	EMP_LNAME	EMP_MANAGER
101	Waddell	102
102	Orincona	
103	Jones	102
104	Reballoh	102
105	Robertson	102
106	Deltona	102

Recursive Relationships

- A **recursive relationship** is one in which a relationship can exist between occurrences of the same entity set. Such a condition is found within a unary relationship.
- For example:
  - The M:N unary relationship may be expressed by "a COURSE may be a prerequisite to many other COURSEs, and each COURSE may have many other COURSEs as prerequisites."



Recursive Relationships

- For example: Implementation of the M:N Recursive Relationship "COURSE requires COURSE"

Table name: COURSE

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

Table name: PREREQ

CRS_CODE	PRE_TAKE
CIS-420	CIS-220
QM-261	MATH-243
QM-362	MATH-243
QM-362	QM-261

References

- *Carlos Coronel and Steven Morris*, "Database Systems: Design, Implementation, and Management" 14th Edition

Questions